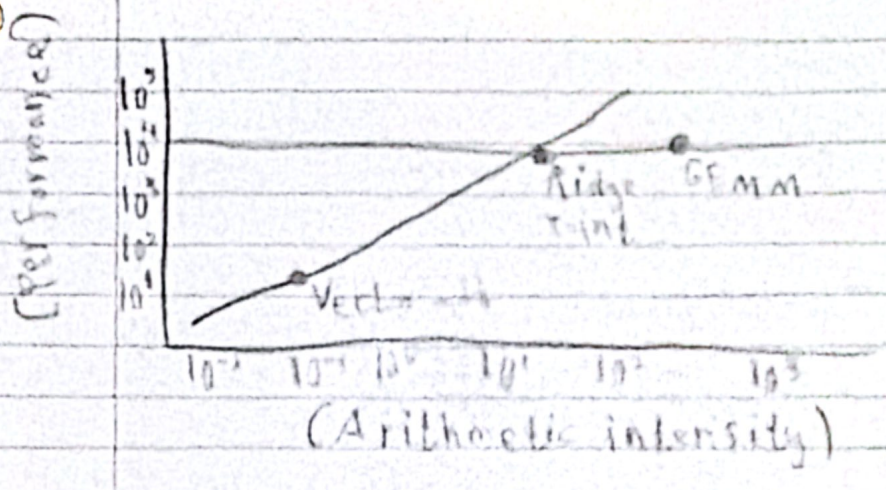
CMAN CS02



Kernel A:

$N = 1024$, FLOPs for GEMM: $2N^3$

FLOPs $= 2 \times 1024^3 = 2{,}147{,}483{,}648$

Bytes transferred $= 3 \times 1024^2 \times 4 = 12{,}582{,}912$ bytes

Arithmetic intensity $= \frac{2{,}147{,}483{,}648}{12{,}582{,}912} = 170.67$ FLOP/byte

Attainable performance:

$P = \min(10{,}000,\ 320 \times 170.67) = 54{,}613.33$

Bound: since $170.67 > 31.25$, GEMM is compute bound

Best architectural change is to increase compute like adding more MAC/FMA unit, because the GEMM is limited by compute not memory

KERNEL B:

$N = 4,194,304$, $C[i] = A[i] + B[i]$

FLOPs $= N$

Bytes transferred:

Read A, B, write C: $3 \times N \times 4 = 50,331,648$ bytes

Arithmetic intensity $= \frac{4,194,304}{50,331,648} = 0.0833$ FLOP/byte

$P = \min(10,000 , 320 \times 0.0833) = 26.67$

Since $0.0833 < 31.25$, vector add is Memory bound

best architectural change is increase DRAM bandwidth, like using wider memory interface.

Because vector add has low AI and is limited by memory traffic